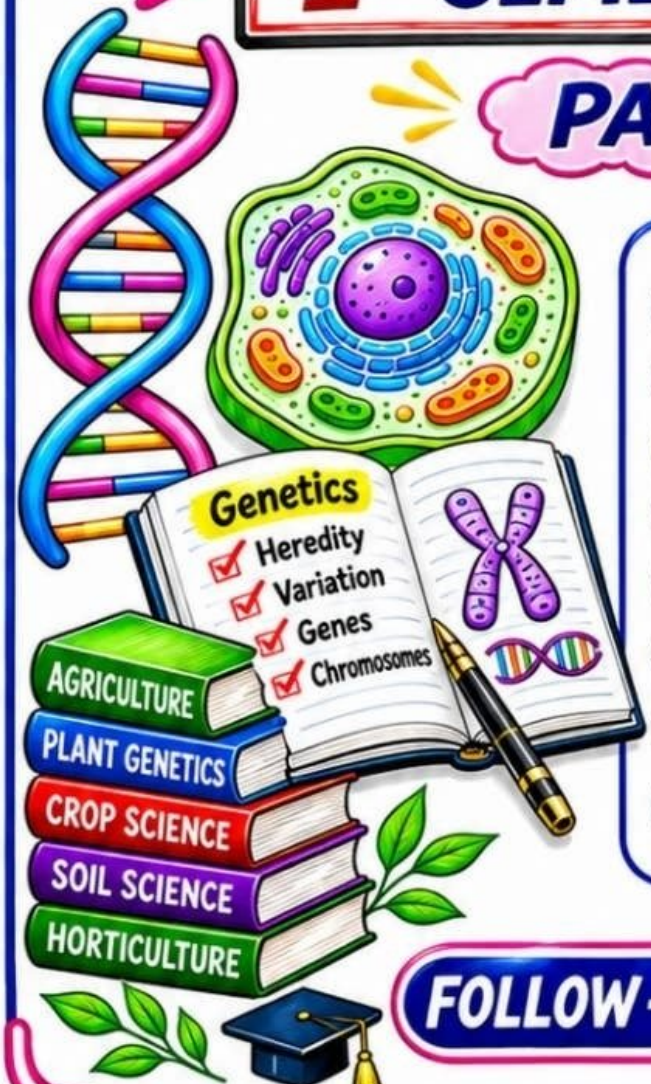
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B.Sc. (Hons.) AGRICULTURE

FUNDAMENTALS OF GENETICS

1st SEMESTER NOTES

PART 1



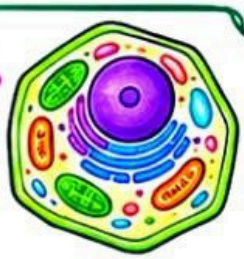
TOPICS COVERED:

1. Cell and Cell Organelles
2. Cell Structure and Functions
3. Plasma Membrane and Cell Wall
4. Endoplasmic Reticulum
5. Ribosomes and Golgi Complex
6. Plastids
7. Nucleus and Nucleolus
8. Chromosome Structure and Classification

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CELL AND CELL ORGANELLES

(Fundamentals of Genetics)



1. Cell

Definition :

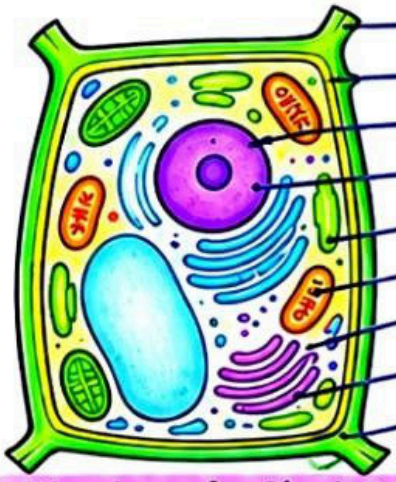
The cell is the basic structural and functional unit of all living organisms. It is the smallest unit capable of performing essential life processes.

Cytology (Cell Biology)

- Cytology is the branch of biology that deals vi'h the study of cell structure, organization and functions.
- It includes the study of different cellular components and their activities.

Cytogenetics

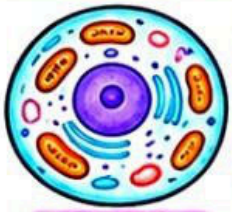
- Cytogenetics is the branch of genetics that studies chromosomes and their role in heredity and variation.
- It combines the principles of Cytology and Genetics.
- It deals vi'h chromosome number, structure, behaviour and abnormalities.



Structure of a Plant Cell



Plant Cell



Animal Cell

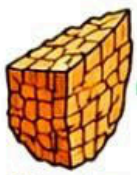
In plant cells, cell wall and chloroplast are present, whereas in animal cells, these are absent.

2. History of Cell Biology

Scientist	Discovery / Contribution
Robert Hooke	Observed cells in cork in 1665 and introduced the term "Cell".
Matthias Schleiden	Proposed that all plants are composed of cells (1838).
Theodor Schwann	Extended cell theory to animals (1839).
Robert Brown	Discovered the nucleus in 1831.
J. E. Purkinje	Introduced the term "Protoplasm" (1839).
Camillo Golgi	Discovered the Golgi apparatus (1898).
Walther Flemming	Described mitosis in detail (1882).
Keith Porter	Helped establish the Endoplasmic Reticulum (1945).
Carl Benda	Introduced the term "Mitochondria" (1898).
Christian de Duve	Discovered lysosomes (1955).

Important Facts

- The word Cell is derived from the Latin word "Cellula", meaning a small room or compartment.
- Robert Hooke observed small box-like compartments in a thin slice of cork.
- Robert Brown discovered the nucleus in plant cells.



Thin sl'ce of cork



Honeycomb-like structure (Cell)



Robert Hooke (1665)

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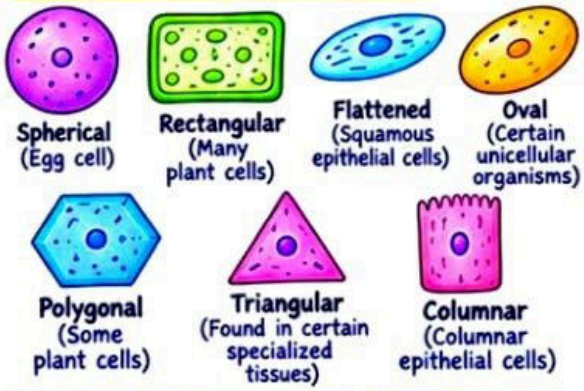
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Cell Shape and Classification

3. Shape and Size of Cells

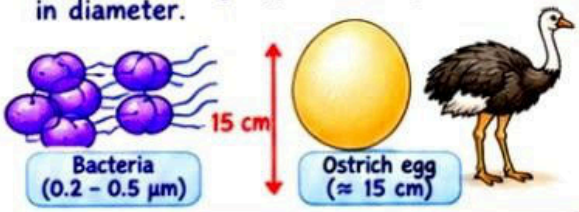
Cells exhibit different shapes depending on their functions.

Common Shapes of Cells



Important Facts About Cell Size

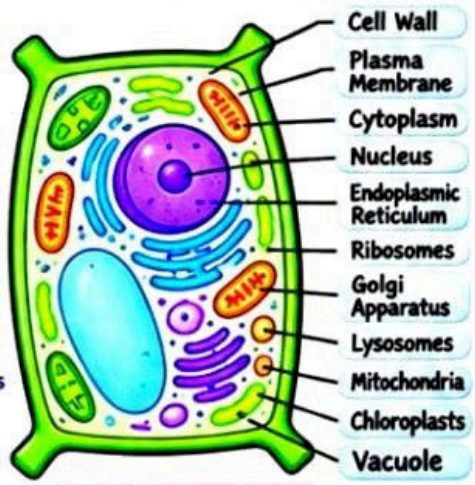
- Many bacteria are extremely small, with some measuring approximately 0.2–0.5 µm.
- The ostrich egg is one of the largest single cells, measuring approximately 15 cm in diameter.



4. General Structure of a Plant Cell

Major Components

1. **Cell Wall** – Provides protection and mechanical support.
2. **Plasma Membrane** – Regulates the movement of substances.
3. **Cytoplasm** – Contains cellular organelles.
4. **Nucleus** – Controls cellular activities and contains genetic material.
5. **Endoplasmic Reticulum** – Involved in synthesis and transport.
6. **Ribosomes** – Sites of protein synthesis.
7. **Golgi Apparatus** – Modifies, sorts and packages cellular products.
8. **Lysosomes** – Contain digestive enzymes.
9. **Mitochondria** – Sites of aerobic respiration and ATP production.
10. **Chloroplasts** – Sites of photosynthesis.
11. **Vacuole** – Helps maintain turgor pressure and stores substances.



Plant Cell (Labelled)

Plant Cell vs Animal Cell

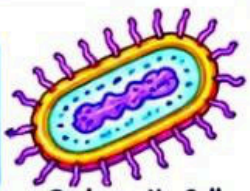
Feature	Plant Cell	Animal Cell
Cell Wall	Present	Absent
Chloroplast	Present in photosynthetic cells	Absent
Vacuole	Usually large central vacuole	Usually small or temporary
Shape	Often fixed or regular	Often variable
Centrioles	Usually absent in higher plants	Usually present

5. Prokaryotic and Eukaryotic Cells

Prokaryotic Cell

A cell that lacks a membrane-bound nucleus and membrane-bound organelles.

Examples: Bacteria and Archaea.



Prokaryotic Cell (Bacteria)

Eukaryotic Cell

A cell that contains a membrane-bound nucleus and membrane-bound organelles.

Examples: Plant cells, animal cells, fungi and protists.



Eukaryotic Cell (Plant/Animal Cell)

Differences Between Prokaryotic and Eukaryotic Cells

Feature	Prokaryotic	Eukaryotic
1. Nuclear Membrane	Absent	Present
2. DNA	Usually circular	Usually linear chromosomes
3. Chromosomes	Usually single main chromosome	Usually multiple chromosomes
4. Nucleolus	Absent	Present
5. Membrane-Bound Organelles	Absent	Present
6. Ribosomes	70S	80S in cytoplasm
7. Cell Wall	Peptidoglycan in bacteria	Cellulose in plants; chitin in fungi
8. Flagella	Structurally different from 9+2 arrangement	Usually 9+2 microtubule arrangement

Note: 70S ribosomes are also found in mitochondria and chloroplasts of eukaryotic cells.

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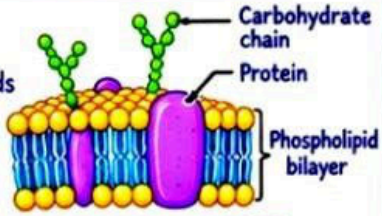
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Plasma Membrane, Cytoplasm and Endoplasmic Reticulum

1. Plasma Membrane / Plasma Lemma

Definition :

The plasma membrane is a thin, living and selectively permeable membrane that surrounds the cytoplasm and separates the cell from its external environment.



Fluid Mosaic Model (Plasma Membrane)

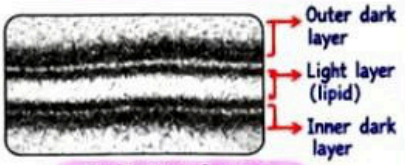
Composition

The plasma membrane mainly consists of:

- Lipids, especially phospholipids.
- Proteins.
- Carbohydrates associated with lipids and proteins.

Structure

- It is mainly composed of a phospholipid bilayer.
- Proteins are embedded in or associated with the lipid bilayer.
- Its structure is explained by the Fluid Mosaic Model, proposed by Singer and Nicolson in 1972.
- Under an electron microscope, it may appear as a trilaminar structure, known as the unit membrane.



Unit Membrane (Trilaminar Structure)

Major Functions

1	Selective Permeability	– Regulates the entry and exit of substances.
2	Transport	– Helps in passive and active transport.
3	Cell Communication	– Contains receptors for receiving signals.
4	Cell Recognition	– Helps cells identify one another.
5	Protection	– Separates the internal environment from the environment.

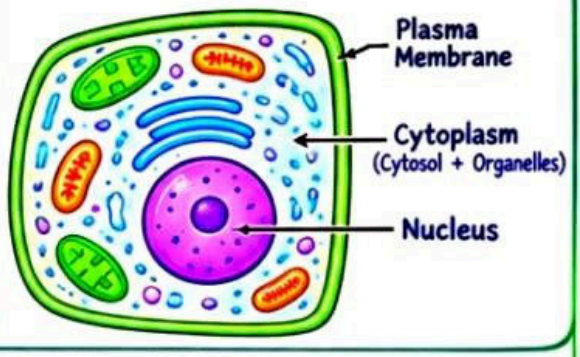
2. Cytoplasm

Definition :

Cytoplasm is the region of the cell between the plasma membrane and the nucleus.

Important Points

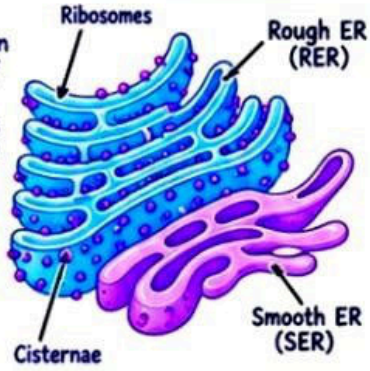
- It contains cytosol and various cell organelles.
- The fluid portion of the cytoplasm is called Cytosol or Hyaloplasm.
- Many important metabolic reactions occur in the cytoplasm.
- Mitochondria and plastids contain their own DNA and ribosomes.
- These organelles are considered semi-autonomous because they can perform some functions independently but still depend on nuclear genes.



3. Endoplasmic Reticulum (ER)

Definition :

Endoplasmic Reticulum is an interconnected network of membrane-bound tubules, vesicles and flattened sacs called cisternae present in the cytoplasm.



Major Components

1. Cisternae
2. Tubules
3. Vesicles

Types of Endoplasmic Reticulum

A. Smooth ER (SER)

- Ribosomes are absent on its surface.
- Involved in lipid and phospholipid synthesis.
- Helps in detoxification of certain harmful substances.
- Participates in carbohydrate metabolism.
- Involved in calcium storage in certain specialized cells.

B. Rough ER (RER)

- Ribosomes are attached to its outer surface.
- Appears rough under an electron microscope.
- Plays an important role in protein synthesis.
- Helps in the processing and transport of newly synthesized proteins.
- Particularly abundant in cells actively producing proteins for secretion or membranes.

Functions of Endoplasmic Reticulum

1 Protein Synthesis and Processing	2 Lipid Synthesis	3 Intracellular Transport	4 Membrane Formation	5 Detoxification
Mainly RER.	Mainly SER.	Helps move substances within the cell.	Contributes to the formation of cellular membranes.	SER helps detoxify certain compounds.

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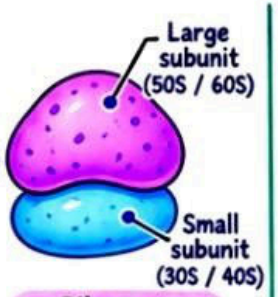
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Ribosomes and Golgi Complex

1. Ribosomes

Definition :

Ribosomes are small, non-membranous structures made of ribosomal RNA (rRNA) and proteins. They are the sites of protein synthesis.



Ribosome (Two Subunits)

Locations of Ribosomes

- Free in the cytoplasm.
- Attached to the Rough Endoplasmic Reticulum.
- In the mitochondrial matrix.
- In chloroplasts.

Types of Ribosomes

1. 70S Ribosomes

- Found in prokaryotic cells.
- Also found in mitochondria and chloroplasts of eukaryotic cells.
- Composed of 50S and 30S subunits.



2. 80S Ribosomes

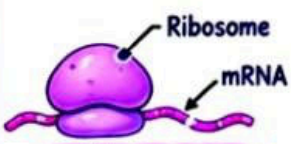
- Found in the cytoplasm of eukaryotic cells.
- Composed of 60S and 40S subunits.



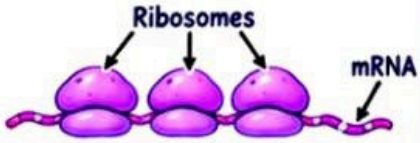
Important : S stands for Svedberg unit, a measure of sedimentation rate. It is not directly additive.

Monosomes and Polysomes

- **Monosome** – A single ribosome translating an mRNA molecule.
- **Polysome / Polyribosome** – A group of ribosomes simultaneously translating the same mRNA molecule.



Monosome (Single Ribosome)



Polysome / Polyribosome (Multiple Ribosomes)

Major Functions

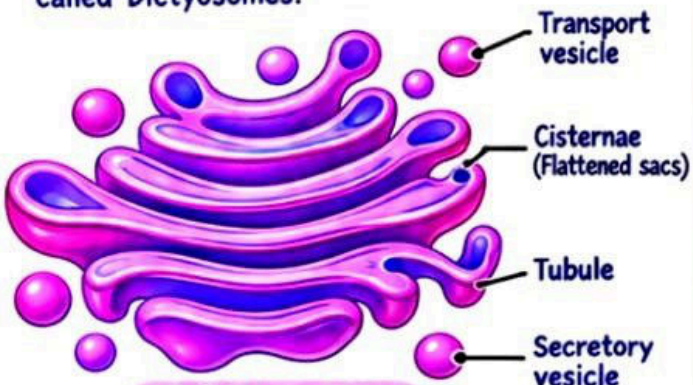
1. Protein synthesis.
2. Formation of peptide bonds between amino acids.
3. Translation of genetic information carried by mRNA into proteins.

2. Golgi Complex / Golgi Apparatus

Definition :

The Golgi apparatus is a membrane-bound organelle involved in modifying, sorting, packaging and transporting proteins and lipids.

- Discovered by Camillo Golgi in 1898.
- In plant cells, Golgi stacks are commonly called Dictyosomes.



Golgi Apparatus (Structure)

Major Components

1. **Cisternae** – Flattened membrane-bound sacs.
2. **Tubules** – Tubular structures associated with the Golgi apparatus.
3. **Vesicles** – Small membrane-bound sacs involved in transport.

Major Functions

1. Modification of proteins and lipids.
2. Sorting and packaging of cellular products.
3. Formation of secretory vesicles.
4. Transport of materials to different cellular destinations.
5. Synthesis and secretion of certain polysaccharides.
6. Helps in cell plate formation during plant cell division.
7. Contributes to the formation of plant cell wall materials such as pectins and hemicelluloses.

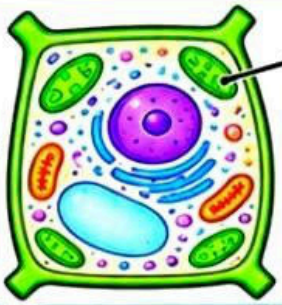
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PLASTIDS

Definition :

Plastids are double-membrane-bound organelles found in plant cells and algae. They are involved in photosynthesis, pigment formation and storage of food materials.



Three Main Types of Plastids

1. Chromoplasts
2. Leucoplasts
3. Chloroplasts

A. Chromoplasts

- These are coloured plastids.
- They contain pigments other than chlorophyll, mainly carotenoids.
- They are responsible for yellow, orange and red colours in many plant parts.



Common Pigments

- Carotene
- Xanthophyll

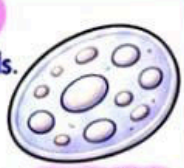


Function :

Provide colour to flowers, fruits and other plant parts, helping attract pollinators and seed dispersers.

B. Leucoplasts

- These are colourless plastids.
- They generally occur in non-photosynthetic storage tissues.
- They are involved in the storage of food materials.



Types of Leucoplasts

Type	Stored Material
Amyloplast	Starch
Elaioplast	Oils / Lipids
Aleuroplast / Proteinoplast	Proteins

Example :

Leucoplasts are commonly found in storage tissues such as roots, tubers and seeds.

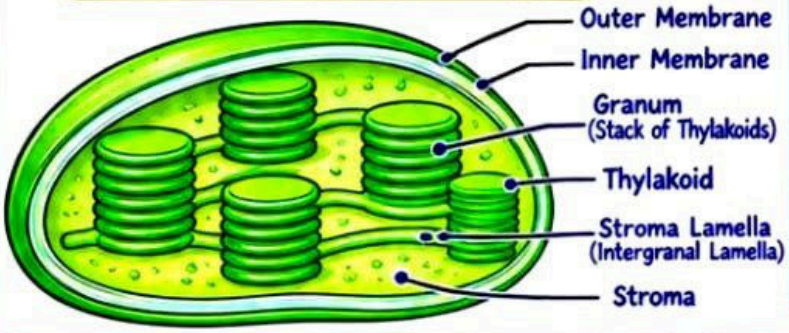
C. Chloroplasts

- Chloroplasts are green plastids containing chlorophyll.
- They are the main sites of photosynthesis.
- They capture light energy and convert it into chemical energy.



Chloroplast

Structure of Chloroplast



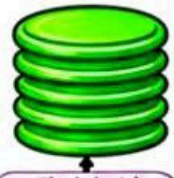
1. **Outer Membrane** – Outer protective membrane.
2. **Inner Membrane** – Inner boundary of the chloroplast.
3. **Stroma** – Fluid-filled internal matrix containing enzymes, DNA and ribosomes.
4. **Thylakoids** – Flattened membrane-bound sacs containing photosynthetic pigments.
5. **Grana** – Stacks of thylakoids.
6. **Stroma Lamellae** – Membranes connecting different grana.

Major Functions of Chloroplasts

1. Photosynthesis.
2. Light absorption by photosynthetic pigments.
3. Formation of ATP and NADPH during the light-dependent reactions.
4. Carbon fixation and carbohydrate synthesis in the stroma.
5. Temporary storage of starch.

Important Facts

- Chlorophyll is the major photosynthetic pigment.
- Chloroplasts contain their own DNA and 70S ribosomes.
- Chlorophyll is located in the thylakoid membranes.



Thylakoid
(with Chlorophyll)

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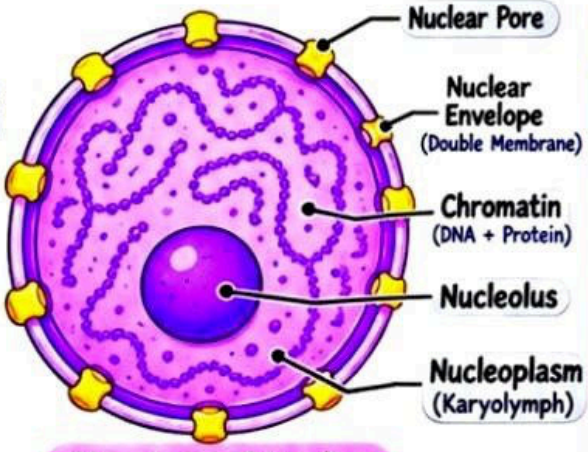
NUCLEUS

Definition :

The nucleus is a membrane-bound organelle that contains most of the genetic material of a eukaryotic cell and regulates cellular activities.

Important Points

- Discovered by Robert Brown in 1831.
- It controls important cellular activities.
- It contains DNA, which carries hereditary information.
- Most eukaryotic cells have a single nucleus, although some cells may be multinucleate or anucleate.



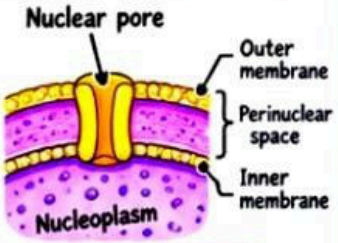
Structure of Nucleus

Major Components of the Nucleus

1. Nuclear Membrane / Nuclear Envelope
2. Nucleoplasm / Karyolymph
3. Chromatin
4. Nucleolus

A. Nuclear Membrane / Nuclear Envelope

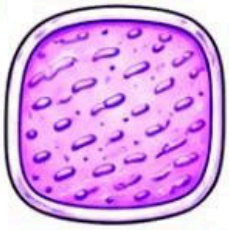
- It is a double-membrane structure surrounding the nucleus.
- It separates the nucleoplasm from the cytoplasm.
- It contains nuclear pores that regulate the exchange of materials.
- The space between the two membranes is called the Perinuclear Space.
- The outer nuclear membrane is continuous with the Endoplasmic Reticulum.



Nuclear Envelope (Enlarged View)

B. Karyolymph / Nucleoplasm

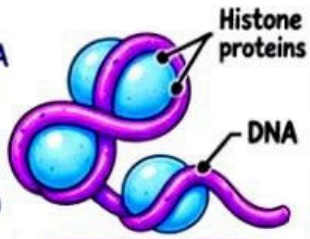
- It is the semi-fluid substance present inside the nucleus.
- It provides a medium for nuclear components.
- It contains proteins, nucleotides, enzymes and other dissolved substances.



Nucleoplasm (Karyolymph)

C. Chromatin

- Chromatin is a complex of DNA and proteins, mainly histones.
- It is present in the nucleus during interphase.
- During cell division, chromatin condenses to form visible chromosomes.



Chromatin (DNA + Histone)

Types of Chromatin

1. Heterochromatin

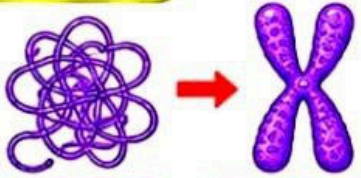
- Highly condensed chromatin.
- Generally transcriptionally less active.
- Stains more darkly.

2. Euchromatin

- Loosely packed chromatin.
- Generally transcriptionally more active.
- Stains more lightly.

Chromatin to Chromosome

- During interphase, DNA exists mainly in the form of chromatin.
- During cell division, chromatin becomes highly condensed and forms chromosomes.



Chromatin (Interphase) → Chromosome (Cell Division)

Chromatin vs Chromosome

Feature	Chromatin	Chromosome
Nature	Loose, thread-like DNA-protein complex	Highly condensed, distinct structure
Stage	Interphase	During cell division
Visibility	Not clearly visible (diffuse)	Clearly visible (under microscope)

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CELL WALL

Cell Wall

Definition :

The cell wall is a rigid, non-living outer covering present outside the plasma membrane of plant cells.

Major Functions

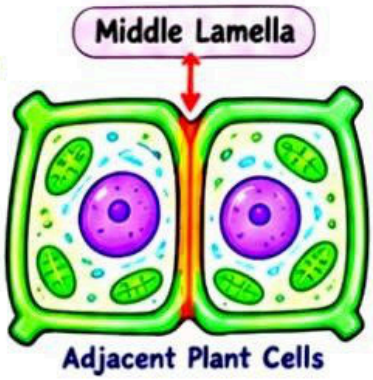
- Provides definite shape to the cell.
- Gives mechanical strength and support.
- Protects the cell from external injury.
- Prevents excessive expansion and helps resist osmotic pressure.
- Provides structural support to plant tissues.

Three Main Layers of the Cell Wall

1. Middle Lamella
2. Primary Cell Wall
3. Secondary Cell Wall

A. Middle Lamella

- It is the cementing layer between adjacent plant cells.
- It helps bind neighbouring cells together.
- It is mainly composed of pectic substances, especially calcium and magnesium pectates.
- It plays an important role in cell-to-cell adhesion.
- It can be degraded by pectinase enzymes.



B. Primary Cell Wall

- It is the first cell wall formed during cell growth.
- It is generally thin, flexible and extensible.
- It allows the cell to expand during growth.

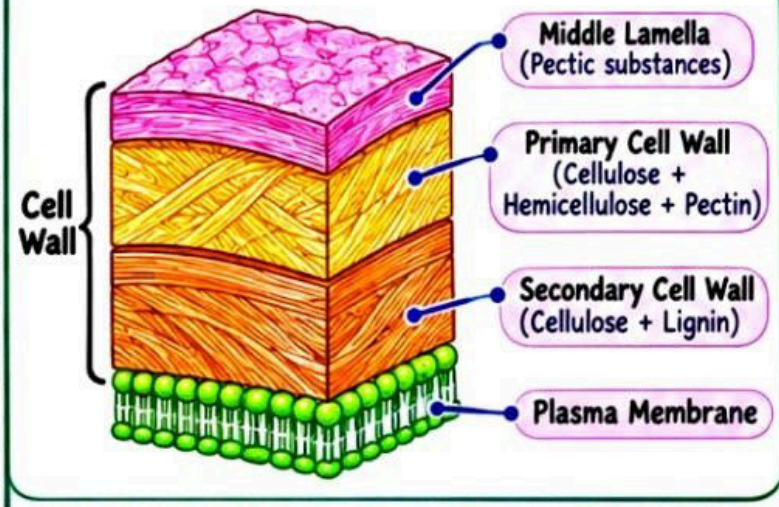
Major Components :

- Cellulose
- Hemicellulose
- Pectin
- Structural proteins

C. Secondary Cell Wall

- It is deposited on the inner side of the primary cell wall after cell expansion has largely stopped.
- It is generally thicker and more rigid than the primary cell wall.
- It commonly contains cellulose and may contain lignin.
- It provides additional mechanical strength.
- It is well developed in many supporting and water-conducting tissues.

Structure of Plant Cell Wall



Important Structural Terms

Cellulose Microfibrils	Fine thread-like structures made of cellulose.
Macrofibrils	Larger fibrillar structures formed by the aggregation of microfibrils.
Lignin	A complex polymer that provides rigidity and strength to many secondary cell walls.

Quick Revision

Middle Lamella	→	Cell Adhesion
Primary Cell Wall	→	Growth and Flexibility
Secondary Cell Wall	→	Strength and Rigidity

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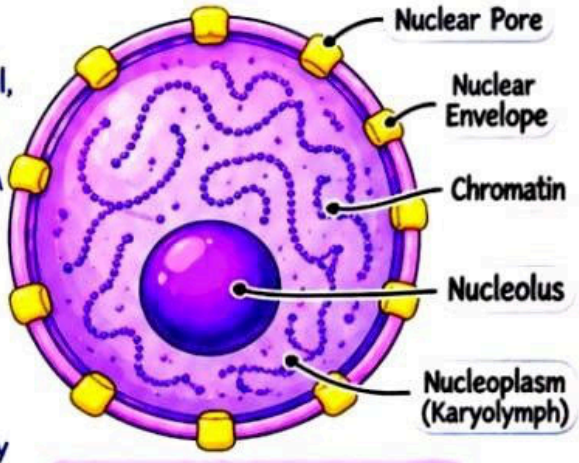
NUCLEOLUS

Definition :

The nucleolus is a dense, spherical, non-membrane-bound structure located inside the nucleus. It is mainly involved in ribosomal RNA (rRNA) synthesis and ribosome assembly.

Location :

- Present inside the nucleus.
- Usually appears as a dense, rounded region.
- Its size and number may vary depending on cell type and metabolic activity.



Nucleus (Showing Nucleolus)

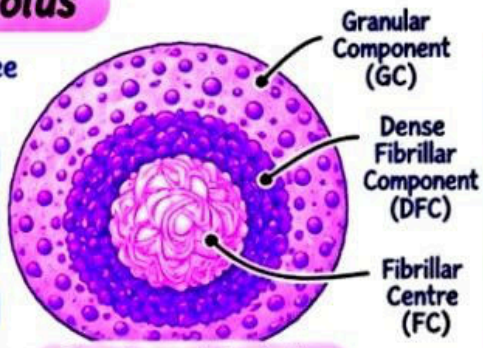
Major Functions

1. Site of ribosomal RNA (rRNA) synthesis.
2. Processing and modification of rRNA.
3. Assembly of ribosomal subunits.
4. Formation of ribosomal subunits that are later transported to the cytoplasm.
5. Associated with nucleolar organizer regions (NORs) present on specific chromosomes.

Structure of Nucleolus

The nucleolus consists of three major components:

1. Fibrillar Centre (FC)
2. Dense Fibrillar Component (DFC)
3. Granular Component (GC)



Structure of Nucleolus (Three Components)

Components and Their Roles

Component	Role / Function
Fibrillar Centre (FC)	• Associated with ribosomal DNA (rDNA) and transcription-related components.
Dense Fibrillar Component (DFC)	• Involved in early processing of newly synthesized rRNA (e.g., cleavage and modification).
Granular Component (GC)	• Associated with later stages of ribosomal subunit assembly (formation of pre-ribosomal particles).

Ribosome Formation (Ribosome Biogenesis)

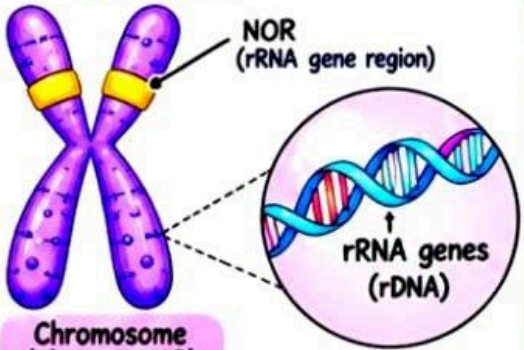
Nucleolus
↓
rRNA Synthesis and Processing
↓
Ribosomal Subunit Assembly
↓
Ribosomal Subunits (Large + Small)
↓
Functional Ribosomes in Cytoplasm

Important Facts

- The nucleolus is located inside the nucleus.
- It is not surrounded by a membrane.
- It is the major site of rRNA production.
- It plays an essential role in ribosome biogenesis (formation of ribosomes).
- Its activity is generally high in cells with active protein synthesis (e.g., meristematic cells, glandular cells).

Nucleolar Organizer Region (NOR)

- NOR is a specific region on certain chromosomes that contains clusters of rRNA genes (rDNA).
- It directs the formation of the nucleolus during interphase.



Chromosome (Showing NOR)

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Chromosome Components and Morphology

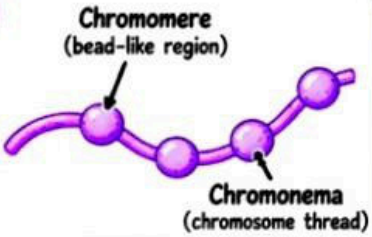
5. Chromomeres

Definition :

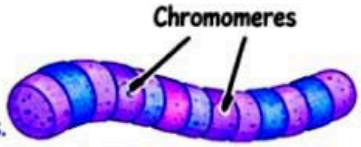
Chromomeres are bead-like condensed regions observed along chromosome threads at certain stages of chromosome organization.

Important Points :

- They appear as localized, densely stained regions.
- They may be visible in extended chromosomes.
- They are especially prominent in some polytene chromosomes.
- Their distribution can help identify chromosome regions.



Chromomere in Chromonema



Polytene Chromosome (Showing Chromomeres)

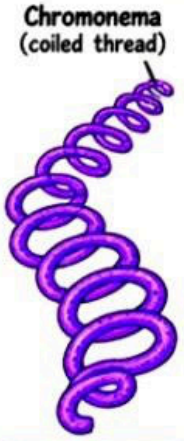
6. Chromonema

Definition :

Chromonema is a classical cytological term for the thread-like coiled structure observed within a chromosome or chromatid.

Important Points :

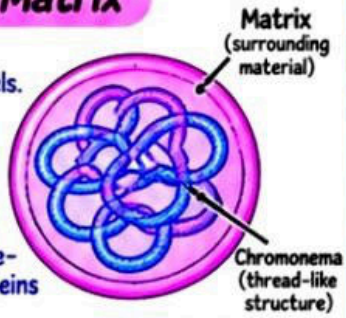
- It describes the thread-like organization of chromosome material.
- It was used in classical models of chromosome structure.
- Modern chromosome organization is explained through DNA, nucleosomes and higher-order chromatin folding.



Chromonema (Enlarged View)

7. Chromosome Matrix

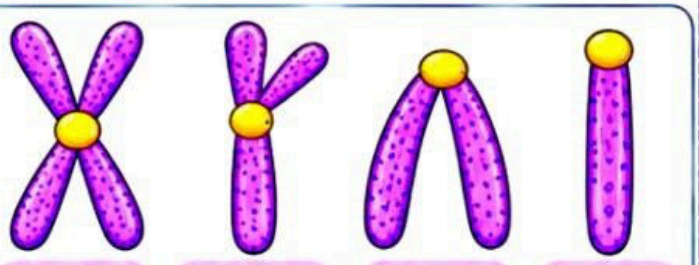
- Matrix is a term used in classical chromosome models.
- It was described as the material surrounding chromosome threads.
- Modern studies emphasize chromatin and chromosome-associated structural proteins rather than a separate, universally defined matrix.



Chromosome with Matrix and Chromonema (Classical Model)

8. Classification Based on Centromere Position

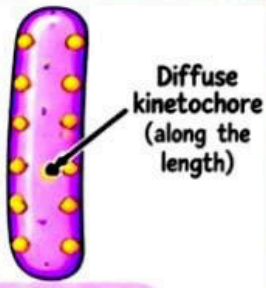
Chromosome Type	Centromere Position	Typical Anaphase Shape
Metacentric	Near the middle (median)	V-shaped
Submetacentric	Away from the middle (sub-median)	L-shaped or J-shaped
Acrocentric	Near one end (sub-terminal)	J-shaped or rod-like
Telocentric	At the terminal end (terminal)	I-shaped or rod-shaped



Metacentric (V-shaped) | Submetacentric (L-shaped) | Acrocentric (J-shaped) | Telocentric (I-shaped)

Additional Type : Holocentric / Holokinetic Chromosomes

- Have distributed kinetochore activity along much of the chromosome.
- Lack a single localized centromere.
- Occur in certain plant and animal groups (eg., some insects, grasses, etc.).



Holocentric Chromosome

Quick Revision

- Meta → V-shaped
- Submeta → L-shaped / J-shaped
- Acro → J-shaped / Rod-shaped
- Telo → I-shaped / Rod-shaped

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Chromosome Structure and Classification

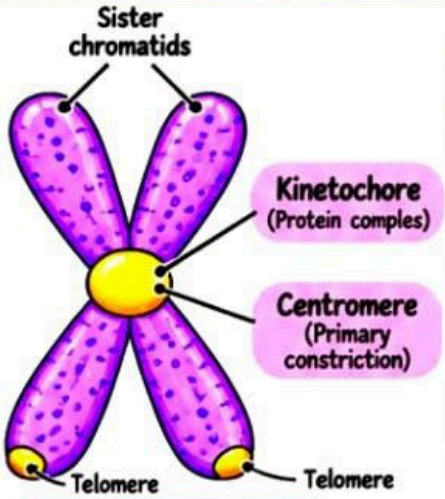
4. Centromere

Definition :

The centromere is the specialized region of a chromosome where the kinetochore assembles and spindle microtubules attach during cell division.

Important Points

- The centromere is associated with the primary constriction of a chromosome.
- It helps ensure proper chromosome segregation.
- The kinetochore is a protein complex assembled at the centromere.
- It plays an important role in chromosome movement during mitosis and meiosis.



Structure of Replicated Chromosome

Functions of the Centromere

1. Provides the site for kinetochore formation.
2. Facilitates attachment of spindle microtubules.
3. Helps proper segregation of chromosomes.
4. Maintains centromeric cohesion between sister chromatids until separation.

5. Chromatids

Definition :

Chromatids are the longitudinal copies of a replicated chromosome.

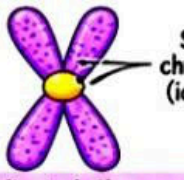
Types of Chromatids

A. Sister Chromatids

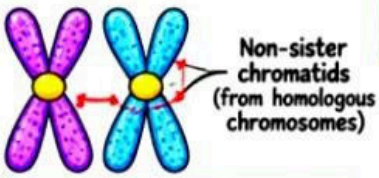
- Formed by DNA replication.
- Joined at the centromeric region until separation.
- Usually contain identical DNA sequences, apart from replication errors or recombination.

B. Non-Sister Chromatids

- Belong to homologous chromosomes.
- May exchange genetic material during crossing over in meiosis I.



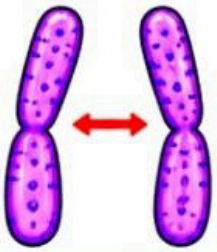
Replicated Chromosome (with Sister Chromatids)



Homologous Chromosomes (Showing Non-sister Chromatids)

Important Facts

- A replicated chromosome consists of two sister chromatids.
- After sister chromatids separate, each is considered an individual chromosome.



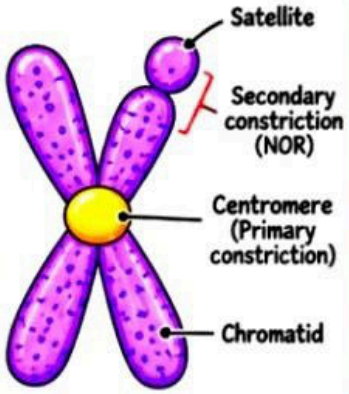
Separation of Sister Chromatids (at Anaphase)

6. Secondary Constriction

- A constricted region other than the primary constriction.
- May be associated with a nucleolar organizer region (NOR).
- NORs contain ribosomal RNA (rRNA) gene clusters.
- Some secondary constrictions separate a satellite from the main chromosome body.

Satellite Chromosome (SAT Chromosome)

A chromosome possessing a small distal segment separated from the main chromosome body by a secondary constriction is called a satellite chromosome.



Satellite Chromosome (SAT Chromosome)

7. Telomere

Definition :

Telomeres are specialized protective structures at the ends of linear chromosomes.

Major Functions

1. Protect chromosome ends from degradation.
2. Reduce inappropriate end-to-end fusion.
3. Help maintain chromosome stability.
4. Participate in the maintenance of chromosome ends during DNA replication.

Important Fact : In most plants, the common telomeric DNA repeat is TTTAGGG.

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Classification of Chromosomes

9. Based on the Number of Centromeres

A. Acentric Chromosome

- A chromosome fragment lacking a functional centromere.
- Generally cannot attach normally to the mitotic spindle.
- May be lost during cell division.

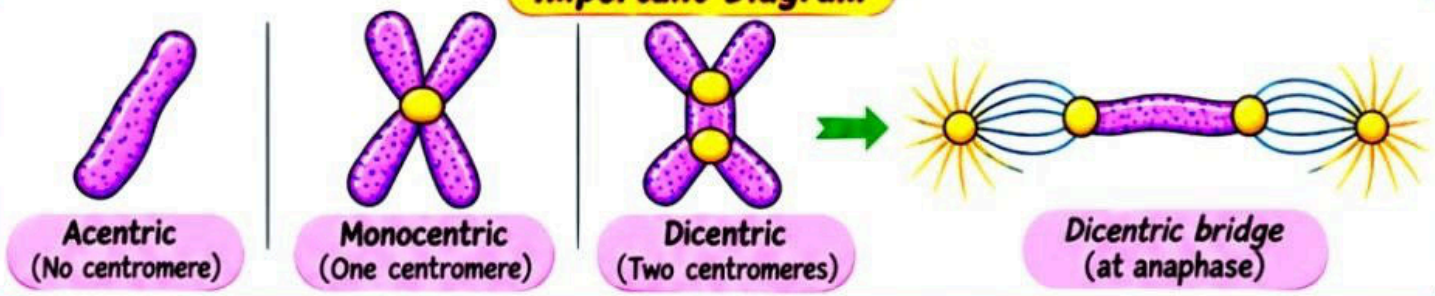
B. Monocentric Chromosome

- Contains one functional centromere.
- Most typical eukaryotic chromosomes are monocentric.
- Ensures normal attachment and segregation during cell division.

C. Dicentric Chromosome

- Contains two centromeres.
- May form following chromosome rearrangements or abnormal chromosome fusion.
- Can form a dicentric bridge during anaphase if both centromeres are pulled toward opposite poles.
- Such bridges may lead to chromosome breakage and genomic instability.

Important Diagram

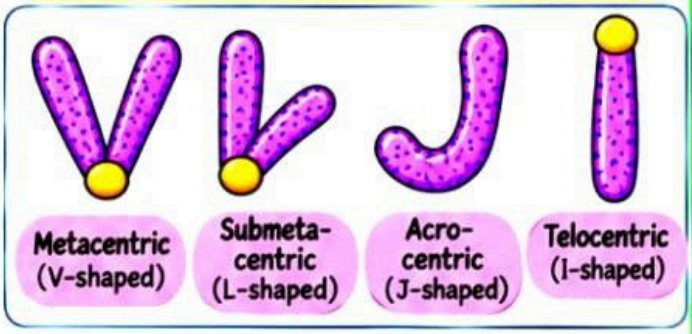


10. Chromosome Shape During Anaphase

Chromosome Type	Centromere Position	Shape at Anaphase
Metacentric	Near the middle (median)	V-shaped
Submetacentric	Away from the middle (sub-median)	L-shaped or J-shaped
Acrocentric	Near one end (sub-terminal)	J-shaped or rod-like
Telocentric	At the terminal end (terminal)	I-shaped / Rod-shaped

Important : Chromosome shape is influenced by centromere position and the relative lengths of chromosome arms.

Anaphase Shapes (Based on Centromere Position)



Exam Revision

- No centromere → Acentric.
- One centromere → Monocentric.
- Two centromeres → Dicentric.
- Distributed centromeric activity → Holocentric.

Memory Trick

Meta	→	V-shaped
Submeta	→	L-shaped / J-shaped
Acro	→	J-shaped / Rod-shaped
Telo	→	I-shaped / Rod-shaped

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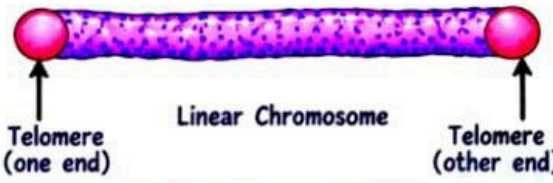
Chromosome Classification

11. Based on Chromosome Structure

A. Linear Chromosomes

- Have two physical ends.
- Typically contain telomeres at both ends.
- Found in the nuclear genome of most eukaryotes.
- DNA is organized into linear chromosomes.

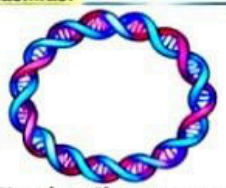
Examples : Plants, animals and fungi.



B. Circular Chromosomes

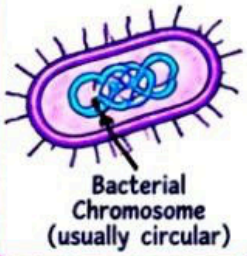
- Have a closed circular DNA structure.
- Common in bacteria and archaea.
- Also found in many plasmids and organellar genomes (e.g., chloroplasts and mitochondria).
- Some viruses contain circular genomes.

Examples : Bacterial chromosomes and circular plasmids.



Important :

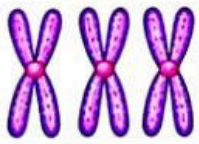
- Not all prokaryotic chromosomes are circular, and not all viral genomes are circular.



12. Based on Essentiality

A. A-Chromosomes

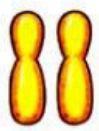
- Constitute the normal chromosome complement of a species.
- Contain genes required for normal growth, development and reproduction.
- Generally show regular inheritance and segregation.



A-Chromosomes
(Normal complement)

B. B-Chromosomes

- Additional chromosomes beyond the normal A-chromosome complement.
- Also called accessory, supernumerary or extra chromosomes.
- Often show irregular or non-Mendelian transmission.
- Usually not essential for normal survival.
- May influence growth, fertility or other traits depending on their number and the species.

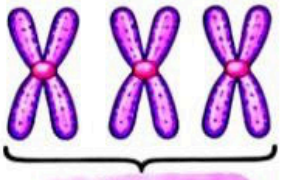


B-Chromosomes
(Additional / Supernumerary)

A-Chromosomes vs B-Chromosomes

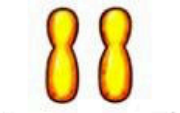
Feature	A-Chromosomes	B-Chromosomes
Occurrence	Normal complement of a species	Additional (extra) chromosomes
Essentiality	Generally essential (for normal life)	Usually non-essential (for survival)
Inheritance	Usually regular (Mendelian)	Often irregular (non-Mendelian)
Number	Typically stable (constant)	May vary (within and between individuals)
Genetic content	Major functional genes	Often repetitive DNA; may contain genes

Diagram : Normal Set + B-Chromosomes



A-Chromosomes
(Normal set)

+



B-Chromosomes
(Additional)

Quick Points :

- A-Chromosomes → Normal, essential, regular inheritance.
- B-Chromosomes → Extra, usually non-essential, irregular inheritance.

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Classification of Chromosomes

13. Based on Role in Sex Determination

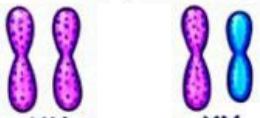
A. Sex Chromosomes (Allosomes)

Definition : Sex chromosomes are chromosomes involved in genetic sex determination in species possessing aıf chromosomal sex-determination system.

Common Systems :

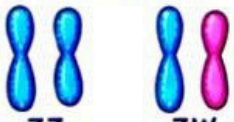
1. XX – XY System

- Female : XX
- Male : XY
- Example : Humans and many mammals.



2. ZZ – ZW System

- Male : ZZ
- Female : ZW
- Example : Birds and many butterflies.

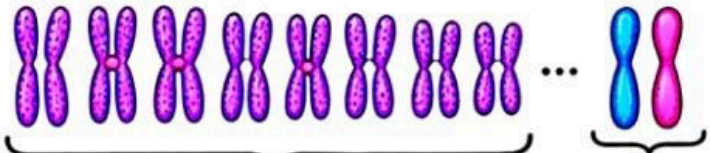


B. Autosomes

- Chromosomes other than the sex chromosomes.
- Generally occur as homologous pairs in diploid organisms.
- Carry genes responsible for numerous structural, physiological and developmental traits.
- Can also carry genes nat influence sexual development.

Example : Humans

- Total chromosomes = 46
- Autosomes = 44 (22 pairs)
- Sex chromosomes = 2 (1 pair)



Important : Sex-determination mechanisms differ among species. Not all organisms use XX–XY or ZZ–ZW systems.

14. Based on Specialized Structure and Function

A. Normal Chromosomes

- Chromosomes forming the usual complement of a species.
- Exhibit the expected structure and segregation pattern.

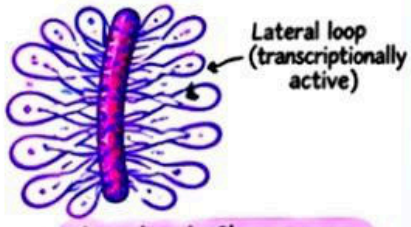
B. Special Chromosomes

- Chromosomes showing distinctive structural of functional characteristics.

Important Examples :

1. Lampbrush Chromosomes

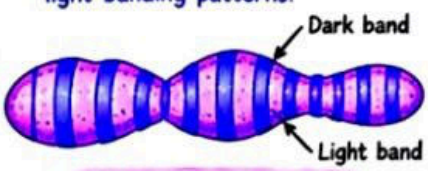
- Large, extended chromosomes with prominent lateral loops.
- Commonly observed during diplotene of meiotic prophase I in growing oocytes of certain animals.
- Associated with intense transcriptional activity.



Lampbrush Chromosome

2. Polytene Chromosomes

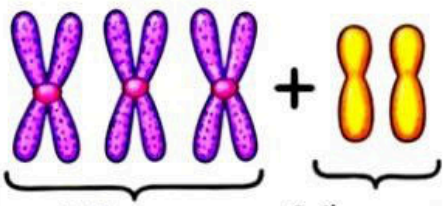
- Giant chromosomes formed through repeated DNA replication without corresponding cell division (endoreplication).
- Contain many aligned DNA copies (sister chromatids).
- Found in certain tissues of dipteran insects, such as salivary glands of *Drosophila*.
- Display characteristic dark and light banding patterns.



Polytene Chromosome
(Banding Pattern)

3. B-Chromosomes

- Additional chromosomes outside the standard chromosome complement.
- Often exhibit unusual inheritance patterns (e.g., non-Mendelian).



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Most Important Questions

Chromosome Structure and Classification

1. Who introduced the term "Chromosome" ?
→ Heinrich Wilhelm Waldeyer, 1888.

2. Who made important early observations of chromosomes during cell division ?
→ Eduard Strasburger and Walther Flemming.

3. At which stage are chromosomes most clearly visible for morphological study ?
→ Metaphase.

4. What is the symbol for haploid chromosome number ?
→ n.

5. What is the symbol for diploid chromosome number ?
→ 2n.

6. What is the symbol for basic chromosome number ?
→ x.

7. What is the basic chromosome number of wheat ?
→ x = 7.

Wheat Chromosome Numbers

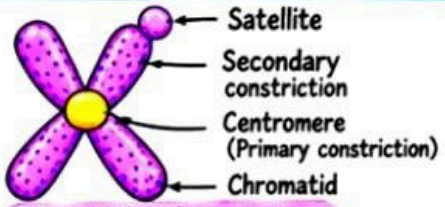
Ploidy Level	Chromosome Number
Diploid (2x)	14
Tetraploid (4x)	28
Hexaploid (6x)	42

Important : Bread wheat (Triticum aestivum) is hexaploid, with 2n = 6x = 42.

8. Where do spindle microtubules attach to chromosomes ?
→ At kinetochores assembled on centromeres.

9. What is another name for the centromeric constriction ?
→ Primary constriction.

10. What is a satellite chromosome ?
→ A chromosome with a small segment separated by a secondary constriction.



Satellite Chromosome

11. What is the typical anaphase shape of a metacentric chromosome ?
→ V-shaped.

12. What is the typical anaphase shape of a telocentric chromosome ?
→ I-shaped or rod-shaped.

13. What is a chromosome with one centromere called ?
→ Monocentric chromosome.

14. What is a chromosome with two centromeres called ?
→ Dicentric chromosome.

15. What is a chromosome fragment without a centromere called ?
→ Acentric fragment.

16. What is the typical structure of nuclear chromosomes in eukaryotes ?
→ Linear DNA chromosomes.

17. What are B-chromosomes also called ?
→ Accessory or supernumerary chromosomes.

18. What is another name for sex chromosomes ?
→ Allosomes.

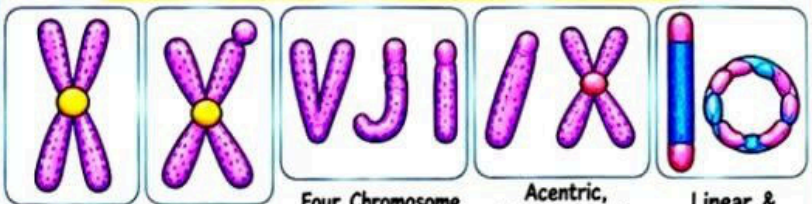
19. What are non-sex chromosomes called ?
→ Autosomes.

20. What is the main characteristic of holocentric chromosomes ?
→ Distributed kinetochore activity along much of the chromosome.

Quick Memory Trick

Meta	→	V-shaped
Submeta	→	L-shaped / J-shaped
Acro	→	J-shaped / Rod-shaped
Telo	→	I-shaped / Rod-shaped

Important Diagrams for Revision


Structure of a Chromosome | Satellite Chromosome | Four Chromosome Types (by Centromere Position) | Acentric, Monocentric & Dicentric Chromosomes | Linear & Circular Chromosomes

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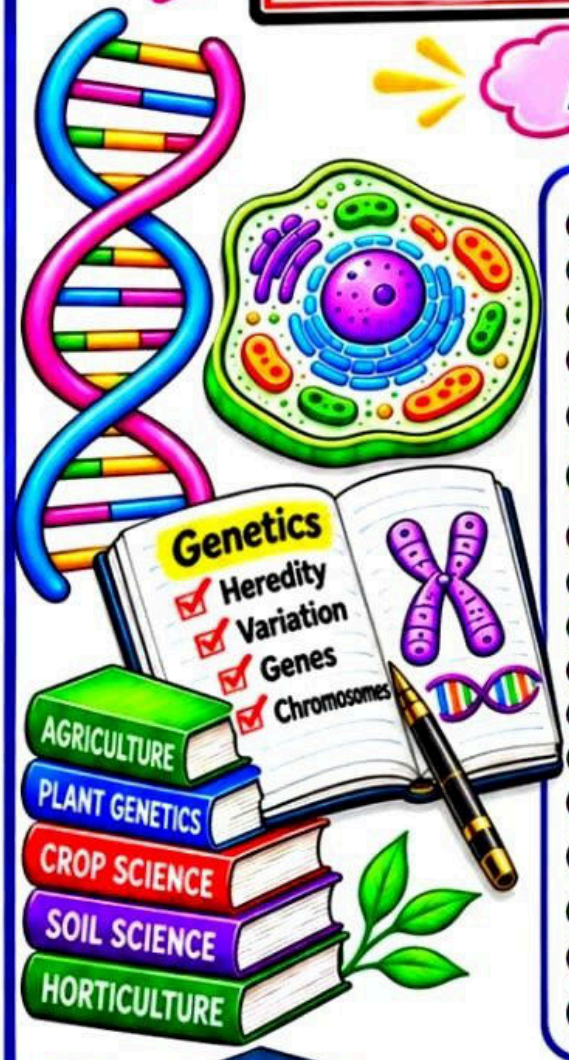
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B.Sc. (Hons.) AGRICULTURE

FUNDAMENTALS OF GENETICS

1st SEMESTER NOTES

PART 2



TOPICS COVERED:

1. Cell Division: Definition, Functions and Types
2. Mitosis: Stages, Characteristics and Significance
3. Meiosis: Stages, Characteristics and Significance
4. Meiosis I: Prophase I and Its Sub-Stages
5. Gregor Johann Mendel: Life, Experiments and Seven Contrasting Traits
6. Factors Responsible for Mendel's Success and Important Genetic Terminology
7. Monohybrid and Dihybrid Crosses
8. Back Cross, Test Cross and Law of Segregation
9. Incomplete Dominance and Codominance
10. Overdominance, Reciprocal Cross and Xenia
11. Gene Interaction and Epistasis: Basic Concepts
12. Types of Epistasis and Non-Epistatic Gene Interaction
13. Dominant Epistasis (12:3:1) and Recessive Epistasis (9:3:4)
14. Duplicate Dominant Epistasis (15:1) and Duplicate Recessive Epistasis (9:7)
15. Dominant and Recessive Epistasis (13:3)
16. Duplicate Genes with Cumulative Effect (9:6:1)
17. Independent Assortment, Poultry Comb Inheritance and Important Genetic Ratios

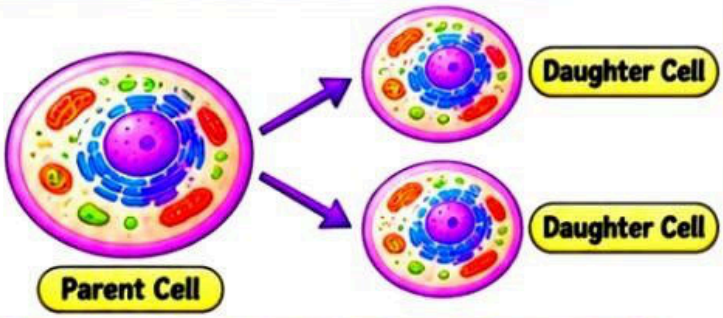
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CELL DIVISION

1. DEFINITION OF CELL DIVISION

- The cell is the **basic structural and functional unit** of all living organisms.
- New cells are produced from **pre-existing cells** through the process of cell division.
- During cell division, the **genetic material** is distributed into daughter cells.

Definition: Cell division is the biological process by which a parent cell divides to form two or more daughter cells.



IMPORTANT TERMS

- **Parent Cell:** The cell that undergoes division.
- **Daughter Cells:** The cells produced after division.

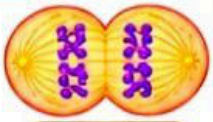
2. FUNCTIONS OF CELL DIVISION

1. Growth and development of the organism.
2. Replacement of damaged or worn-out cells.
3. Repair and regeneration of tissues.
4. Reproduction in unicellular organisms.
5. Maintenance of chromosome number across generations.

3. TWO MAIN EVENTS OF CELL DIVISION

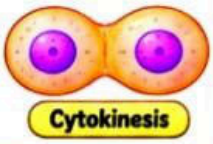
A. KARYOKINESIS

- Division of the nucleus and its genetic material into daughter nuclei.



B. CYTOKINESIS

- Division of the cytoplasm to form separate daughter cells.

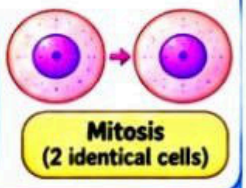


Karyokinesis + Cytokinesis = Cell Division

4. TYPES OF CELL DIVISION

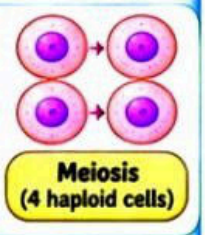
A. MITOSIS

- Also called **equational division.**
- Usually produces **two daughter cells.**
- Daughter cells generally retain the same chromosome number as the parent cell.



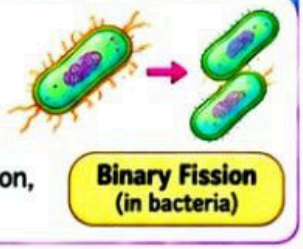
B. MEIOSIS

- Also called **reduction division** because chromosome number is reduced by half.
- Involves two successive nuclear divisions.
- Usually produces **four haploid cells.**



C. BINARY FISSION

- Common in bacteria.
- A parent cell divides into two daughter cells.
- It is a form of asexual reproduction, not mitosis.



COMPARISON OF MITOSIS AND MEIOSIS

Feature	Mitosis	Meiosis
Name	Equational division	Reduction division
Number of divisions	One	Two (Meiosis I & II)
Number of daughter cells	Two	Four
Chromosome number in daughter cells	Same as parent (2n)	Half of parent (n)
Role	Growth, repair, replacement	Formation of gametes and genetic variation

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MITOSIS

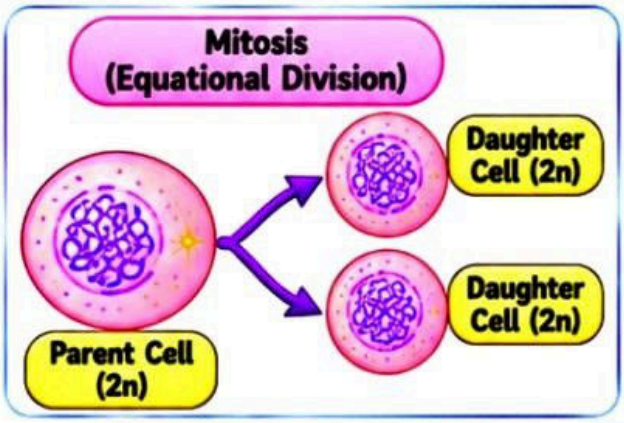
1. DEFINITION OF MITOSIS

- The term Mitosis was introduced by **Walther Flemming** in **1882**.
- Mitosis is a type of cell division in which **one parent cell** produces **two genetically similar daughter cells**.
- It generally occurs in **somatic cells**.
- The chromosome number remains **unchanged**.

Example: A diploid parent cell (2n) produces two diploid daughter cells (2n).

2. MAJOR CHARACTERISTICS OF MITOSIS

1. One nuclear division takes place.
2. Two daughter cells are generally formed.
3. Daughter cells usually have the same chromosome number as the parent cell.
4. It is known as **equational division**.
5. It supports growth, development and tissue repair.



3. STAGES OF THE CELL CYCLE

- The cell cycle consists of:

1. Interphase
2. M Phase (Mitotic Phase)

A. INTERPHASE

- Interphase is the period between two successive cell divisions. It is the **longest phase** of the cell cycle in many actively dividing cells.

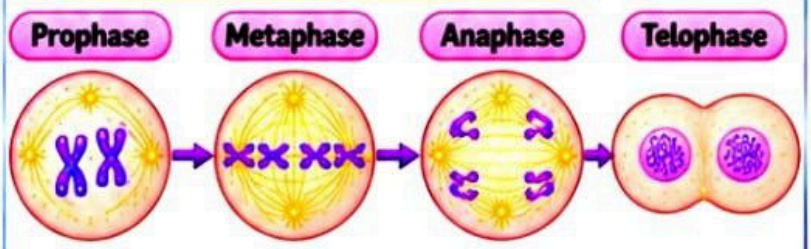
THREE MAIN STAGES OF INTERPHASE

Stage	Major Activities
G_1 Phase	Cell growth; synthesis of RNA and proteins.
S Phase	DNA replication and chromosome duplication.
G_2 Phase	Further growth; synthesis of proteins required for mitosis.

IMPORTANT FACTS

- DNA replication occurs during the **S phase**.
- Each replicated chromosome consists of **two sister chromatids**.
- Sister chromatids remain joined at the **centromeric region** until their separation.
- Some cells may enter a resting or quiescent stage called G_0.

4. STAGES OF MITOSIS



Sequence: P → M → A → T

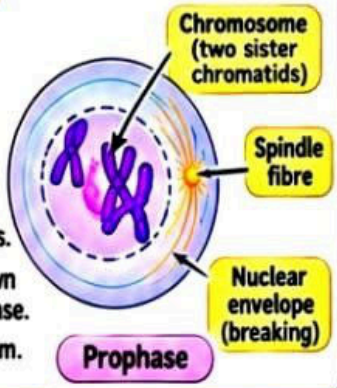
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STAGES OF MITOSIS

5. PROPHASE

Major Events:

1. Chromatin condenses to form visible **chromosomes**.
2. Each chromosome consists of **two sister chromatids**.
3. The **nucleolus** gradually disappears.
4. The **nuclear envelope** breaks down during late prophase / prometaphase.
5. The **mitotic spindle** begins to form.

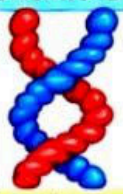


RELATIONAL COILING

- The coiling of sister chromatids around each other is traditionally described as relational coiling.

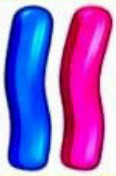
Two Classical Forms:

Plectonemic Coiling



Sister chromatids intertwined (coiled around each other)

Paranemic Coiling

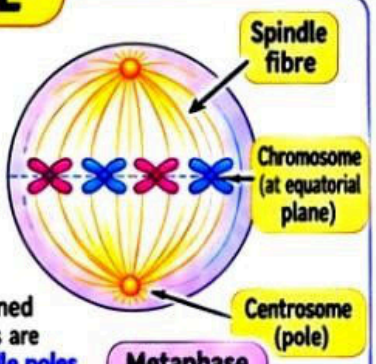


Sister chromatids lie alongside each other (without tight interlocking)

6. METAPHASE

Major Events:

1. Chromosomes become **highly condensed**.
2. Spindle microtubules attach to **kinetochores**.
3. Chromosomes align at the **equatorial plane**.
4. Each chromosome is positioned so that its sister kinetochores are connected to **opposite spindle poles**.

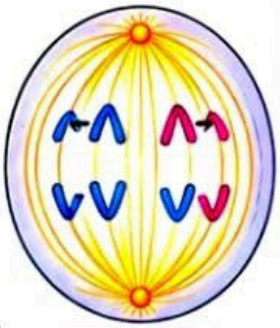


Important Term: The equatorial plane is also called the metaphase plate.

7. ANAPHASE

Major Events:

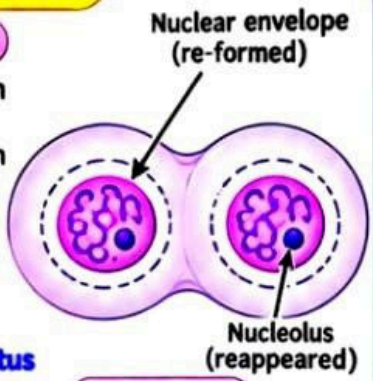
1. **Cohesion** between sister chromatids is released.
2. **Centromeric regions** separate.
3. Sister chromatids move towards **opposite poles**.
4. Each separated chromatid is now considered an **individual chromosome**.
5. **Spindle microtubules** help move chromosomes towards the poles.



8. TELOPHASE

Major Events:

1. Chromosomes reach **opposite poles**.
2. Chromosomes begin to **decondense**.
3. **Nuclear envelopes** re-form around each chromosome set.
4. **Nucleoli** reappear.
5. The **spindle apparatus** disassembles.



RESULT: Two daughter nuclei are formed.

QUICK REVISION

Stage	Key Event / Feature
Prophase	Chromosome condensation.
Metaphase	Chromosome alignment at the equatorial plane (metaphase plate).
Anaphase	Separation of sister chromatids and movement to opposite poles.
Telophase	Re-formation of nuclear envelope and decondensation of chromosomes.

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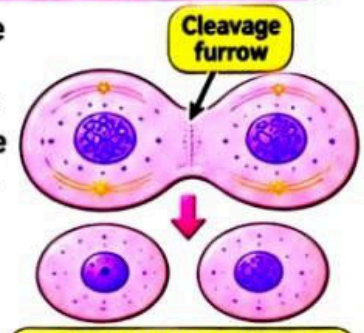
CYTOKINESIS AND MEIOSIS

9. CYTOKINESIS

Definition: Cytokinesis is the division of the cytoplasm following nuclear division, resulting in separate daughter cells.

CYTOKINESIS IN ANIMAL CELLS

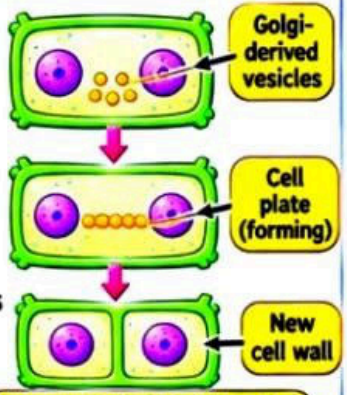
- Occurs through the formation of a **cleavage furrow**.
- The cell membrane constricts inward.
- The cytoplasm divides into two daughter cells.



Animal Cell Cytokinesis

CYTOKINESIS IN PLANT CELLS

- Occurs through the formation of a **cell plate**.
- Golgi-derived vesicles accumulate at the equatorial region.
- The vesicles fuse to form the cell plate.
- The cell plate develops into the new cell wall separating the daughter cells.



Plant Cell Cytokinesis

10. RESULT OF MITOSIS

- One parent cell generally produces two daughter cells.

In a diploid cell: $2n \rightarrow 2n + 2n$

- Both daughter cells generally have the same chromosome number.
- They are genetically similar to each other and the parent cell, except for mutations or other genic changes.

11. MEIOSIS

Definition: Meiosis is a specialized type of cell division in which one diploid cell undergoes two successive nuclear divisions to produce haploid cells.

IMPORTANT CHARACTERISTICS

1. It involves two successive divisions: **Meiosis I** and **Meiosis II**.
2. DNA replication occurs **once** before Meiosis I.
3. **No DNA replication** occurs between Meiiosisl and Meiosis II.
4. Homologous chromosomes separate during Meiosis I.
5. Sister chromatids separate during Meiosis II.
6. It reduces the chromosome number from diploid **(2n)** to haploid **(n)**.

TWO SUCCESSIVE DIVISIONS

Meiosis I (Reductional Division)

$2n \rightarrow n + n$

Reduction of chromosome number to half.

Meiosis II (Equational Division)

$n + n \rightarrow$ **Four haploid cells (n, n, n, n)**

Separation of sister chromatids without change in chromosome number.

12. PRE-MEIOTIC INTERPHASE

- DNA replication occurs during the **S phase** before Meiosis I.
- Each chromosome becomes composed of **two sister chromatids**.
- The cell prepares for meiotic division.

IMPORTANT: DNA replication does **not** occur between Meiosis I and Meiosis II.

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MEIOSIS – DETAILED STAGES

13. MEIOSIS I

- Meiosis I is called **Reductional Division** because homologous chromosomes separate, reducing the chromosome number by half.

MAJOR EVENTS OF MEIOSIS I

1. Pairing of homologous chromosomes.
2. Crossing over during Prophase I.
3. Separation of homologous chromosomes during Anaphase I.
4. Reduction of chromosome number from 2n to n.

14. PROPHASE I

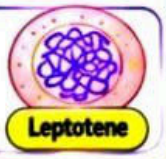
- Prophase I is the longest and most complex stage of meiosis. It is divided into five substages.

SEQUENCE:

Leptotene → Zygotene → Pachytene → Diplotene → Diakinesis

14.1 LEPTOTENE

- Chromosomes begin to condense.
- They appear as long, thin threads.
- Chromomeres may be visible along the chromosomes.



14.2 ZYGOTENE

- Homologous chromosomes begin pairing.
- This pairing is called synapsis.
- The synaptonemal complex develops between homologs.



14.3 PACHYTENE

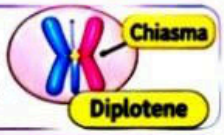
- Synapsis is complete.
- Homologous chromosomes form bivalents.
- Each bivalent contains four chromatids and is called a tetrad.
- Crossing over occurs between non-sister chromatids.



EXAM FACT: Crossing over occurs during Pachytene.

14.4 DIPLOTENE

- Homologous chromosomes begin to separate.
- They remain attached at certain points called chiasmata.



14.5 DIAKINESIS

- Chromosomes become highly condensed.
- Chiasmata undergo terminalization.
- The nucleolus disappears.
- The nuclear envelope breaks down.
- The spindle apparatus develops.



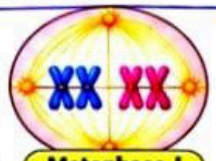
FINAL RESULT OF MEIOSIS

- One diploid parent cell generally produces four haploid products.

2n → 4n
(Four haploid cells)

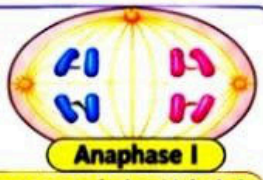
15. METAPHASE I

- Bivalents arrange themselves at the equatorial plane.
- Homologous chromosomes are oriented towards opposite poles.
- Spindle microtubules attach to kinetochores.



16. ANAPHASE I

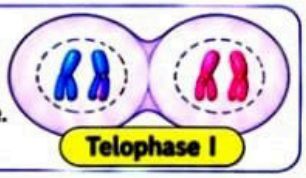
- Homologous chromosomes separate.
- Each chromosome moves towards an opposite pole.
- Centromeres do not divide.
- Sister chromatids remain joined.



IMPORTANT: Reduction in chromosome number occurs during Meiosis I.

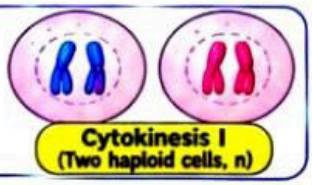
17. TELOPHASE I

- Homologous chromosomes reach opposite poles.
- Nuclear envelopes may re-form.
- Chromosomes may partially decondense.
- Two haploid nuclei may be formed.



18. CYTOKINESIS I

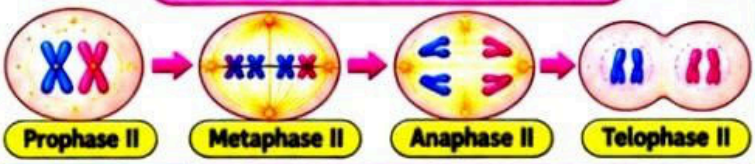
- Cytoplasm divides.
- Two haploid daughter cells are generally formed.
- Each chromosome still consists of two sister chromatids.



19. MEIOSIS II

- Meiosis II is called **Equational Division** because sister chromatids separate without further reducing the chromosome number.

STAGES OF MEIOSIS II



Stage	Major Events
Prophase II	Chromosomes condense again; nuclear envelope breaks down (if re-formed); spindle apparatus develops.
Metaphase II	Chromosomes align at the equatorial plane; spindle microtubules attach to kinetochores.
Anaphase II	Centromeres divide; sister chromatids separate; chromosomes move towards opposite poles.
Telophase II	Chromosomes reach the poles; nuclear envelopes re-form; chromosomes decondense.

IMPORTANT EXAM FACTS

- Synapsis → **Zygotene**
- Crossing over → **Pachytene**
- Chiasmata become visible → **Diplotene**
- Terminalization → **Diakinesis**
- Homologous chromosome separation → **Anaphase I**
- Sister chromatid separation → **Anaphase II**

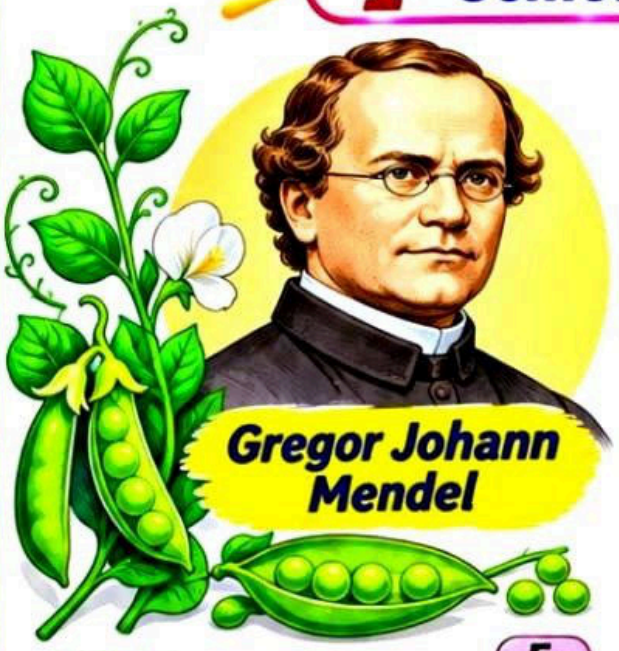
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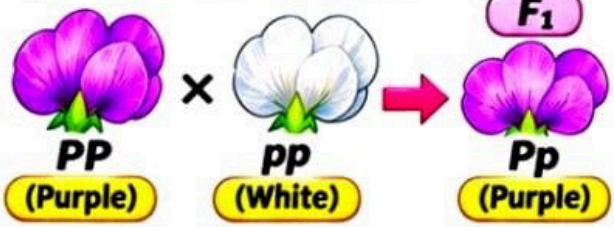
B.Sc. (Hons.) AGRICULTURE

FUNDAMENTALS OF GENETICS

PART 4 – MENDEL'S LAWS OF INHERITANCE

1st Semester | Study Notes





TOPICS COVERED

1. Introduction to Gregor Johann Mendel
2. Selection of Pea Plant
3. Seven Contrasting Characters
4. Mendel's Experimental Work
5. Important Genetic Terminology
6. Monohybrid Cross
7. Dihybrid Cross
8. Back Cross and Test Cross
9. Law of Segregation

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Learn • Understand • Revise • Succeed

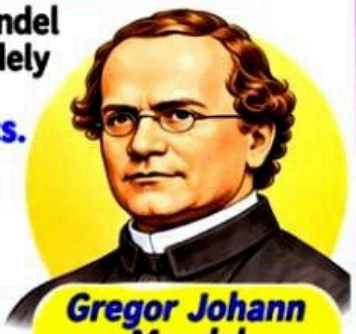
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MENDEL'S LAWS OF INHERITANCE

1. INTRODUCTION TO GREGOR JOHANN MENDEL

- Gregor Johann Mendel (1822–1884) is widely known as the **Father of Genetics.**





Gregor Johann Mendel (1822 – 1884)

IMPORTANT FACTS ABOUT MENDEL

- Born on **20 July 1822** in Heinzendorf, Austrian Empire (now in the Czech Republic).
- He was an **Augustinian monk** and scientist.
- He conducted his famous experiments on garden pea plants (**Pisum sativum**).
- His experiments the carfrged out at the Augustinian monastery in **Brno**.
- He presented his findings in **1865**.
- His research paper, titled **Experiments on Plant Hybridization**, was published in **1866**.
- His work was independently rediscovered around **1900** by **Hugo de Vries, Carl Correns** and **Erich von Tschermak**.

2. WHY DID MENDEL SELECT THE PEA PLANT?

Mendel selected the garden pea (**Pisum sativum**) because:

1. It has many easily distinguishable contrasting characters.
2. It is naturally self-pollinated.
3. Artificial cross-pollination can be performed easily.
4. It has a relatively short generation time.
5. It produces a large number of seeds.
6. True-breeding varieties are available.
7. It is easy to grow and maintain.



Garden Pea (Pisum sativum)

3. SEVEN PAIRS OF CONTRASTING CHARACTERS STUDIED BY MENDEL

	Character	Dominant Trait	Recessive Trait
1	Seed Shape	Round	Wrinkled
2	Seed Colour	Yellow	Green
3	Flower Colour	Purple (Violet)	White
4	Pod Shape	Inflated	Constricted
5	Pod Colour	Green	Yellow
6	Flower Position	Axial	Terminal
7	Stem Height	Tall	Dwarf

4. WHY WERE MENDEL'S RESULTS SUCCESSFUL?

1. He studied one or a few characters at a time.
2. He selected clear contrasting traits.
3. He used true-breeding parental lines.
4. He maintained accurate records of large numbers of plants.
5. He applied mathematical and statistical analysis.
6. He repeated experiments across generations.

IMPORTANT NOTE: Mendel's experiments established the basic principles of inheritance, although some traits in nature do not follow simple Mendelian ratios.

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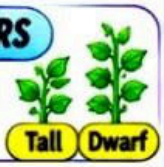
IMPORTANT GENETIC TERMINOLOGY

5. BASIC TERMS IN GENETICS

1 HEREDITARY FACTOR (GENE)
A unit of heredity that influences a particular character.



2 CONTRASTING CHARACTERS
Alternative forms of a character, such as tall and dwarf plants.



3 DOMINANT ALLELE
An allele that expresses its effect in a heterozygote.

A

4 RECESSIVE ALLELE
An allele whose effect is masked by a dominant allele in a heterozygote.

a

5 HOMOZYGOUS
An individual having two identical alleles at a particular gene locus.

AA / aa

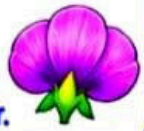
Examples: AA or aa.

6 HETEROZYGOUS
An individual having two different alleles at a particular gene locus.

Aa

Example: Aa.

7 PHENOTYPE
The observable characteristics of an organism.



Examples: Tall, dwarf, purple flower.

8 GENOTYPE
The genetic constitution of an organism.

TT / Tt / tt

Examples: TT, Tt or tt.



Garden Pea (*Pisum sativum*)

6. MONOHYBRID CROSS

Definition: A monohybrid cross is a genetic cross involving one pair of contrasting characters.





EXAMPLE: PLANT HEIGHT

Parental Generation (P)	TT (Tall) × tt (Dwarf)
F_1 Generation	All Tt – Tall
F_1 Selfing	Tt × Tt
F_2 Genotypes	TT, Tt, Tt, tt

Genotypic Ratio	Phenotypic Ratio
1 TT : 2 Tt : 1 tt	3 Tall : 1 Dwarf

7. DIHYBRID CROSS

Definition: A dihybrid cross is a genetic cross involving two pairs of contrasting characters.

EXAMPLE: SEED SHAPE AND SEED COLOUR

R = Round | r = Wrinkled | Y = Yellow | y = Green

Parental Generation (P)	RRYY × rryy
F_1 Generation	All RrYy – Round and Yellow
Gametes produced by F_1	RY, Ry, rY, ry
F_1 Selfing	RrYy × RrYy

F_2 Phenotypic Ratio		
	9	Round Yellow
	3	Round Green
	3	Wrinkled Yellow
	1	Wrinkled Green

F_2 Phenotypic Ratio = 9 : 3 : 3 : 1

F_2 Genotypic Ratio

1 RRYY : 2 RRYy : 1 RRyy :
2 RrYY : 4 RrYy : 2 Rryy :
1 rrYY : 2 rrYy : 1 rryy

IMPORTANT FACT: The 9 : 3 : 3 : 1 ratio is expected when the two gene pairs show independent assortment and complete dominance, with no gene interaction affecting the traits.

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BACK CROSS, TEST CROSS & MENDEL'S LAW OF SEGREGATION

8. BACK CROSS

Definition: A back cross is a cross between an F_1 hybrid and either of its original parents.

EXAMPLES

1. F_1 Hybrid (Tt) × Dominant Parent (TT)
2. F_1 Hybrid (Tt) × Recessive Parent (tt)

IMPORTANT POINT: Every test cross is type of back cross, but every back cross is not a test cross.

9. TEST CROSS

Definition: A test cross is a cross between an individual showing a dominant phenotype and a homozygous recessive individual to determine the unknown genotype.

EXAMPLE: Tall plant (T_) × Dwarf plant (tt)

CASE 1: UNKNOWN PLANT IS HOMOZYGOUS DOMINANT

TT × tt → All offspring: Tt (Tall)

CASE 2: UNKNOWN PLANT IS HETEROZYGOUS

Tt × tt → Offspring: 1 Tt (Tall) : 1 tt (Dwarf)

Phenotypic Ratio = 1 : 1

IMPORTANT USE: A test cross helps determine whether an individual with a dominant phenotype is homozygous or heterozygous.

MENDEL'S FIRST LAW – LAW OF SEGREGATION

Also known as the Law of Purity of Gametes.

10. STATEMENT OF THE LAW

The two alleles of a gene pair separate from each other during gamete formation, so each gamete receives only one allele of the pair.

EXAMPLE

Parental Cross (P)	RR × rr
F_1 Generation	All Rr
Gametes produced by F_1	R and r
F_1 Selfing	Rr × Rr
F_2 Genotypes	RR, Rr, Rr, rr

Genotypic Ratio	Phenotypic Ratio
1 : 2 : 1	3 Dominant : 1 Recessive

EASY WAY TO REMEMBER

- ✅ Alleles remain distinct; they do not blend.
- ✅ The two alleles separate during gamete formation.
- ✅ Each gamete receives only one allele of a gene pair.
- ✅ Fertilization restores the paired condition.

IMPORTANT EXAM FACTS

✅ Monohybrid phenotypic ratio	3 : 1
✅ Monohybrid genotypic ratio	1 : 2 : 1
✅ Dihybrid phenotypic ratio	9 : 3 : 3 : 1
✅ Monohybrid test cross ratio (heterozygote)	1 : 1
✅ Dihybrid test cross ratio (double heterozygote, independent assortment)	1 : 1 : 1 : 1
✅ Law of Segregation	Separation of alleles during gamete formation

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INCOMPLETE DOMINANCE AND CODOMINANCE

1. INCOMPLETE DOMINANCE

Definition: In incomplete dominance, neither allele is completely dominant over the other. The heterozygote shows an **intermediate** phenotype between the two homozygous parents.

EXAMPLE: FLOWER COLOUR IN MIRABILIS JALAPA



Red (RR) | Pink (Rr) | White (rr)

Parental Generation (P)	Red Flower (RR) × White Flower (rr)
F_1 Generation	All Rr – Pink Flowers
F_1 Selfing	Rr × Rr
F_2 Genotypic Ratio	1 RR : 2 Rr : 1 rr
F_2 Phenotypic Ratio	1 Red : 2 Pink : 1 White

KEY POINTS

- ✅ The heterozygote has an intermediate phenotype.
- ✅ Neither allele completely masks the other.
- ✅ The F_2 phenotypic ratio is **1 : 2 : 1**.

2. CODOMINANCE

Definition: Codominance occurs when **both** alleles in a heterozygote are fully and simultaneously expressed.

EXAMPLE: ABO BLOOD GROUP SYSTEM

- The A and B alleles are codominant.
- Both A and B antigens are expressed together in the heterozygote (AB blood group).

Cross: I^AI^A (Blood Group A) × I^BI^B (Blood Group B)

↓

F_1 Generation: All I^AI^B – Blood Group AB

AB Blood Group (I^AI^B)



Both A and B antigens are expressed on the red blood cells.

IMPORTANT ABO BLOOD GROUP GENOTYPES

Blood Group	Genotype
A	I^AI^A or I^Ai
B	I^BI^B or I^Bi
AB	I^AI^B
O	ii



INCOMPLETE DOMINANCE VS CODOMINANCE

Feature	Incomplete Dominance	Codominance
Heterozygous phenotype	Intermediate	Both traits expressed
Example	Pink flowers in Mirabilis jalapa	AB blood group
F_2 ratio in a typical monohybrid cross	1 : 2 : 1	1 : 2 : 1

REMEMBER: *Incomplete dominance produces an intermediate phenotype, whereas codominance expresses both alleles together.*

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IMPORTANT GENETIC CONCEPTS

3. OVERDOMINANCE

Definition: Overdominance occurs when the heterozygote has a higher fitness or a more pronounced phenotype than either homozygote.

- The heterozygote shows an advantage over both homozygous conditions.
- It is also associated with heterozygote advantage.



Heterozygote Advantage

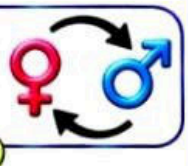
IMPORTANT DISTINCTION: If the heterozygote shows an intermediate phenotype between two homozygotes, the phenomenon is incomplete dominance, not overdominance.

4. RECIPROCAL CROSS

Definition: A reciprocal cross involves two crosses in which the sexes of the parental genotypes are reversed.

EXAMPLE

Cross 1:	Female A × Male B
Cross 2:	Female B × Male A

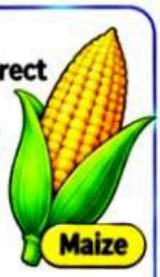


IMPORTANCE

1. It helps investigote the role of parental sex in inheritance.
2. Similar results in both crosses generally suggest that the trait is controlled by nuclear genes.
3. Different results may indicate maternal, cytoplasmic or sex-linked effects, depending on the trait.

5. XENIA

Definition: Xenia refers to the direct effect of pollen genotype on the characteristics of the developing seed or endosperm in the same generation.



Maize

EXAMPLE: MAIZE

- Pollen from a genetically different maize plant may influence the colour or other characteristics of the endosperm.
- This effect can be observed in the seeds produced in the same generation.

IMPORTANT: Xenia is associated with the direct effect of the pollen genotype on the endosperm or seed tissues.

6. LAW OF INDEPENDENT ASSORTMENT

Statement: During gamete formation, alleles of different gene pairs assort independently of the another, provided the genes are unlinked or sufficiently far apart on the same chromosome.

EXAMPLE: DIHYBRID CROSS

Parental Generation (P)	TTYY × ttyy
F_1 Generation	All TtYy
Gametes produced by F_1	TY, Ty, tY, ty
F_1 Selfing	TtYy × TtYy
F_2 Phenotypic Ratio	9 : 3 : 3 : 1

- 9 Tall Yellow
- 3 Tall Green
- 3 Dwarf Yellow
- 1 Dwarf Green

IMPORTANT EXAM FACT: The law of independent assortment is generally applicable to genes located on different chromosomes or genes that are sufficiently far apart on the same chromosome. Linked genes may not show independent assortment.

IMPORTANT GENETIC RATIOS

	Cross / Inheritance Pattern	Genotypic Ratio	Phenotypic Ratio
1	Monohybrid (complete dominance)	1 : 2 : 1	3 : 1
2	Incomplete dominance	1 : 2 : 1	1 : 2 : 1
3	Codominance	1 : 2 : 1	1 : 2 : 1
4	Dihybrid (independent assortment)	1:2:1:2:4:2:1:2:1	9 : 3 : 3 : 1
5	Monohybrid test cross (heterozygote)	1 : 1	1 : 1

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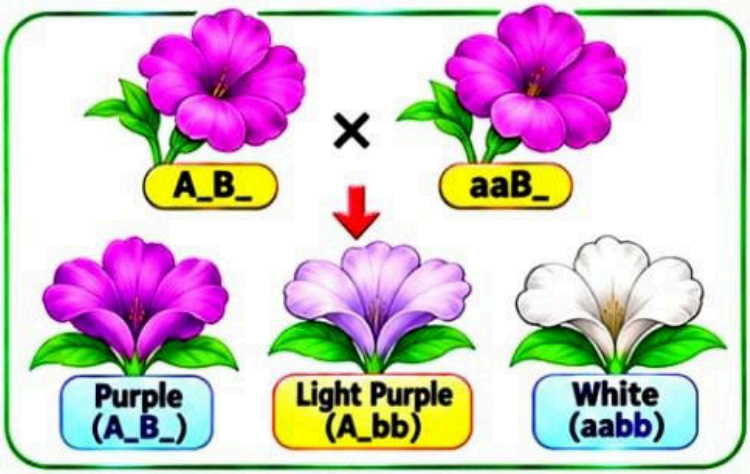
FUNDAMENTALS OF GENETICS

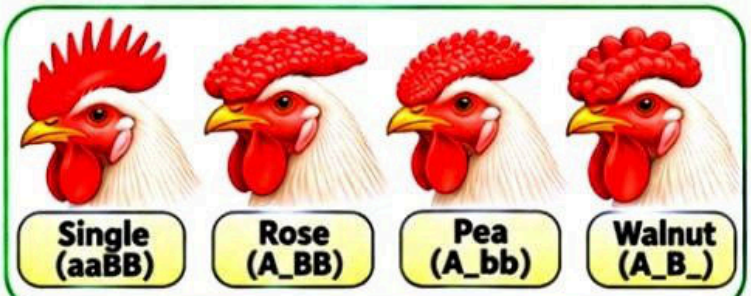
CHAPTER 5 – EPISTASIS AND GENE INTERACTION

Study Notes | 1st Semester

TOPICS COVERED

- Gene Interaction
- Epistasis and Its Types
- Dominant Epistasis
- Recessive Epistasis
- Duplicate Dominant Epistasis
- Duplicate Recessive Epistasis
- Dominant and Recessive Epistasis
- Duplicate Genes with Cumulative Effect
- Poultry Comb Inheritance
- Important Genetic Ratios





Epistasis Ratio (Examples)

- Dominant epistasis → 12 : 3 : 1
- Recessive epistasis → 9 : 3 : 4
- Duplicate dominant → 15 : 1
- Duplicate recessive → 9 : 7
- Dominant & recessive → 13 : 3
- Cumulative effect → 9 : 6 : 1

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CHAPTER 5

EPISTASIS AND GENE INTERACTION

1. WHAT IS GENE INTERACTION?

Definition: Gene interaction occurs when **two or more genes** influence the expression of **a single character.**

- Different genes may work together to control one character.
- Their combined effects may produce different phenotypes.
- Gene interaction can modify the typical Mendelian phenotypic ratios.

TYPES OF GENE INTERACTION

1. Intergenic Interaction
2. Non-allelic Gene Interaction



✓ These terms describe interactions between genes located at different loci.

2. WHAT IS EPISTASIS?

Definition: Epistasis is a type of gene interaction in which an allele of one gene **masks or modifies** the expression of another **non-allelic gene.**

IMPORTANT TERMS

Epistatic Gene:	The gene that masks or modifies the expression of another gene.
Hypostatic Gene:	The gene whose expression is masked or modified by the epistatic gene.

EXAMPLE

- If Gene A masks the expression of Gene B:
  - Gene A = Epistatic gene
  - Gene B = Hypostatic gene



3. DIFFERENCE BETWEEN DOMINANCE AND EPISTASIS

Feature	Dominance	Epistasis
Interaction	Between alleles of the same gene	Between different genes
Location	Same gene locus	Different gene loci
Effect	One allele masks another allele	One gene masks or modifies another gene
Example	T masks t (Dominant allele T over recessive t)	An inhibitor gene masks another gene's expression (e.g., colour inhibitor gene in plants)

EASY WAY TO REMEMBER

✓ **Dominance:** Interaction between alleles at the same locus.

✓ **Epistasis:** Interaction between genes at different loci.



IMPORTANT

Epistasis involves **non-allelic genes** and may **modify** the expected Mendelian phenotypic ratios.

TYPES OF EPISTASIS

4. MAJOR TYPES OF GENE INTERACTION

Type of Interaction	Typical F_2 Phenotypic Ratio
Non-epistatic interaction	9 : 3 : 3 : 1
Dominant epistasis	12 : 3 : 1
Recessive epistasis	9 : 3 : 4
Duplicate dominant epistasis	15 : 1
Duplicate recessive / Complementary genes	9 : 7
Dominant and recessive epistasis	13 : 3
Duplicate genes with cumulative effect	9 : 6 : 1

5. DOMINANT EPISTASIS – 12 : 3 : 1

Definition: In dominant epistasis, a dominant allele of one gene masks the expression of another gene, regardless of the alleles present at the second locus.

EXAMPLE: GRAIN COLOUR IN SORGHUM

Consider two genes:

- W = Dominant inhibitor allele
- w = Recessive allele
- Z = Allele associated with grain colour
- z = Alternative allele
- The dominant W allele masks the expression of the Z gene.



Sorghum

GENOTYPE – PHENOTYPE RELATIONSHIP

Genotype	Phenotype
W_Z_	Red
W_zz	Red
wwZ_	Pearly
wwzz	Chalky



Red | Pearly | Chalky

F_2 PHENOTYPIC RATIO

12 Red : 3 Pearly : 1 Chalky

REMEMBER:

- Dominant Epistasis = 12 : 3 : 1
- A dominant allele at one locus masks the expression of another gene.

6. NON-EPISTATIC INTERACTION – 9 : 3 : 3 : 1

- In non-epistatic interaction, two gene pairs independently influence two aspects of a character without one gene masking the other.

EXAMPLE: POULTRY COMB SHAPE

Two genes control comb shape:

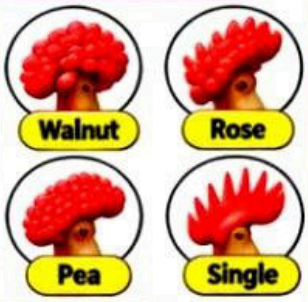
- R = Rose comb allele
- r = Recessive allele
- P = Pea comb allele
- p = Recessive allele



Poultry Comb

GENOTYPE – PHENOTYPE RELATIONSHIP

Genotype	Comb Type
R_P_	Walnut
R_pp	Rose
rrP_	Pea
rrpp	Single



Walnut | Rose | Pea | Single

F_2 PHENOTYPIC RATIO

9 Walnut : 3 Rose : 3 Pea : 1 Single

IMPORTANT EXAM FACT: The poultry comb example demonstrates interaction between two non-allelic genes and gives the classical 9 : 3 : 3 : 1 ratio.

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RECESSIVE EPISTASIS – 9 : 3 : 4

7. DEFINITION

Recessive epistasis occurs when the **homozygous recessive condition** of one gene **masks the expression** of another gene at a different locus.

Example: *The genotype **aa** masks the expression of gene **B**.*

GENERAL GENOTYPE – PHENOTYPE PATTERN

Genotype	Phenotype
A_B_	Phenotype 1
A_bb	Phenotype 2
aaB_	Phenotype 3
aabb	Phenotype 3

Two classes with the genotype **aa** are grouped together.

F_2 PHENOTYPIC RATIO

9 : 3 : 4

8. EXAMPLE: SEED COAT COLOUR IN PEA

In this illustrative model:

- A_B_ → Yellow
- A_bb → Green
- aaB_ → White
- aabb → White



F_2 PHENOTYPIC RATIO

9 Yellow : 3 Green : 4 White

ANOTHER EXAMPLE: BODY COLOUR IN DROSOPHILA

Genotype	Phenotype
A_B_	Grey
A_bb	Black
aaB_	Yellow
aabb	Yellow



F_2 PHENOTYPIC RATIO

9 Grey : 3 Black : 4 Yellow

KEY POINTS FOR EXAMINATION

- Recessive epistasis occurs when a homozygous recessive genotype at one locus masks another gene.
- The masking genotype is **aa** in the examples above.
- Two phenotypic classes are combined.
- The typical F_2 ratio is **9 : 3 : 4.**



REMEMBER: Recessive Epistasis = 9 : 3 : 4

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GENE INTERACTION – IMPORTANT RATIOS

9. DUPLICATE RECESSIVE EPISTASIS – 9 : 7

Also Known As:
Complementary Gene Interaction.

Definition: In duplicate recessive interaction, dominant alleles of **both genes** are required together to produce one phenotype. The absence of a dominant allele at either locus results in the **same alternative phenotype.**

KEY CONCEPT

- Both dominant alleles (A and B) are required for Phenotype 1.
- If either A or B is absent (homozygous recessive), Phenotype2 is produced.

GENERAL GENOTYPE – PHENOTYPE PATTERN

Genotype	Phenotype
A_B_	Phenotype 1
A_bb	Phenotype 2
aaB_	Phenotype 2
aabb	Phenotype 2

F_2 PHENOTYPIC RATIO
9 : 7

EXAMPLE: GRAIN COLOUR IN SORGHUM

- B_1 and B_2 are two dominant genes.
- Both dominant alleles are required for brown grain colour.
- If either dominant gene is absent, the grain is white.


Brown


White

Genotype	Phenotype
B_1_$B2$_	Brown
B_1_bb_2	White
bb_1B_2_	White
bb_1bb_2	White

F_2 PHENOTYPIC RATIO
9 Brown : 7 White

REMEMBER: Complementary Genes = 9 : 7

10. DOMINANT AND RECESSIVE EPISTASIS – 13 : 3

Also Known As:
Inhibitory Gene Interaction (in a common classical model).

Definition: In this interaction, a dominant inhibitor allele at one locus **masks the expression of another gene.** A particular recessive genotype at the second locus may also produce the same phenotype.

GENOTYPE – PHENOTYPE RELATIONSHIP

Genotype	Phenotype
I_C_	No pigment
I_cc	No pigment
iiC_	Pigmented
iicc	No pigment


No Pigment


Pigmented

F_2 PHENOTYPIC RATIO
13 No pigment : 3 Pigmented

GENES IN THE MODEL

- I = Dominant inhibitor allele
- i = Non-inhibitor allele
- C = Allele required for pigment production
- c = Recessive allele

IMPORTANT:



- The 13 : 3 ratio is associated with a specific inhibitory interaction model.
- Do not confuse it with the ordinary 9 : 3 : 3 : 1 independent assortment ratio.

REMEMBER: Dominant and Recessive Epistasis = 13 : 3

DUPLICATE GENES WITH CUMULATIVE EFFECT – 9:6:1

11. DEFINITION

Duplicate genes with cumulative effect involve **two non-allelic genes** whose dominant alleles contribute together to produce a stronger or distinct phenotype.

- When both dominant genes are present, one phenotype is produced.
- When only one dominant gene is present, a second phenotype is produced.
- When both genes are homozygous recessive, a third phenotype is produced.

GENERAL GENOTYPE – PHENOTYPE PATTERN

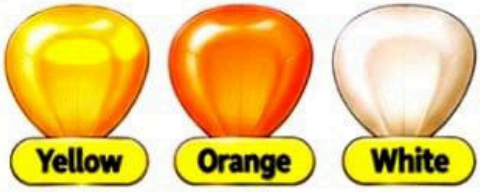
Genotype	Phenotype
A_B_	Phenotype 1
A_bb or aaB_	Phenotype 2
aabb	Phenotype 3

F_2 PHENOTYPIC RATIO

9:6:1

EXAMPLE: KERNEL COLOUR IN MAIZE

- Both dominant genes present → Yellow
- Only one dominant gene present → Orange
- Both genes recessive → White



F_2 PHENOTYPIC RATIO

9 Yellow : 6 Orange : 1 White

KEY POINTS

- ✓ Two non-allelic genes contribute to the same character.
- ✓ Both dominant genes together produce one phenotype.
- ✓ A single dominant gene produces another phenotype.
- ✓ Both recessive genes produce a third phenotype.
- ✓ The typical F_2 ratio is **9:6:1**.

REMEMBER

- Both dominant genes → Phenotype 1
- One dominant gene → Phenotype 2
- Both recessive genes → Phenotype 3

Duplicate Genes with Cumulative Effect = 9:6:1

12. QUICK REVISION – IMPORTANT EPISTASIS RATIOS

Type of Gene Interaction	F_2 Ratio
Non-epistatic interaction	9:3:3:1
Dominant epistasis	12:3:1
Recessive epistasis	9:3:4
Duplicate dominant epistasis	15:1
Duplicate recessive / Complementary genes	9:7
Dominant and recessive epistasis	13:3
Duplicate genes with cumulative effect	9:6:1

13. IMPORTANT TERMS

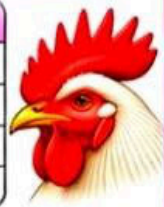
Epistatic Gene:	A gene that masks or modifies the expression of another non-allelic gene.
Hypostatic Gene:	A gene whose expression is masked or modified by another gene.
Gene Interaction:	The combined influence of two or more genes on a character.
Non-epistatic Interaction:	Interaction in which neither gene masks the expression of the other.

14. IMPORTANT EXAMPOINTS

1. Dominant epistasis → 12:3:1
2. Recessive epistasis → 9:3:4
3. Duplicate dominant epistasis → 15:1
4. Duplicate recessive interaction → 9:7
5. Dominant and recessive epistasis → 13:3
6. Duplicate cumulative interaction → 9:6:1
7. Poultry comb inheritance → 9:3:3:1

15. POULTRY COMB – QUICK REVISION

Genotype	Comb Type
R_P_	Walnut
R_pp	Rose
rrP_	Pea
rrpp	Single



F_2 RATIO:

9 Walnut : 3 Rose : 3 Pea : 1 Single

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